# Practice Exam 2 · Short Version

Half length: four parts, **50 points, 50 minutes**. Calculator and one A4 sheet allowed. Show your working. Short answers: one to three sentences. Suggested time: A 5 min · B 17 min · C 15 min · D 10 min · 3 min to check.

**Scenario for Parts B to D.** A gym chain wants to contact members who are likely to cancel within 60 days and offer them a free month. The offer is the action. A missed canceller costs **4,000**. An unnecessary offer costs **500**. All money is in fictional units.

Formula sheet: Accuracy (TP+TN)/N · Sensitivity TP/(TP+FN) · Specificity TN/(TN+FP) · Precision TP/(TP+FP) · Total cost C_FN×FN + C_FP×FP · Break-even p* = C_FP/(C_FP+C_FN) · z = b0 + b1·x1 + b2·x2, p = e^z/(1+e^z) · odds ratio = e^b. "Flagged" means the model recommends the offer.

## Part A · Concept checks (6 points: items 1–4 multiple choice, 1 point each; item 5 true/false with a one-sentence reason, 2 points)

**1.** Adding the field *retention_offer_sent* (an offer staff send after a member phones to cancel) raises test AUC from 0.74 to 0.97. What should the chain do?

(A) Deploy: 0.97 is excellent.
(B) Lower the threshold to use the extra signal.
(C) Check when each input becomes known, and drop inputs that exist only after a member has started to cancel.
(D) Collect more rows so the effect is estimated more precisely.

**2.** The team trained on a 50/50 resampled file. AUC is unchanged, but the average predicted probability is 0.45 while the real cancellation rate is 12%. What should they do before applying the break-even threshold?

(A) Nothing: AUC did not change.
(B) Use 0.5 as the threshold because the data are balanced.
(C) Remove the members who cancelled.
(D) Recalibrate the probabilities to the real base rate on untouched recent data, or set the threshold by rank.

**3.** Lowering the threshold from 0.50 to 0.20 will usually:

(A) raise recall, lower precision and flag more members.
(B) raise both recall and precision.
(C) lower recall and raise precision.
(D) change nothing, because AUC stays the same.

**4.** An analyst tests 30 candidate variables at α = 0.05 and reports the 2 that are significant. If none truly matters, roughly how likely is at least one false positive?

(A) 5%
(B) about 79%
(C) 30%
(D) 2%

**5.** True or false, with one sentence: if an input has p < 0.001, changing that input will change cancellations.

## Part B · Reading regression models (17 points)

A logistic regression predicts whether a member cancels within 60 days.

| Term | Coefficient | Std. error | p-value |
|---|---|---|---|
| Intercept | −1.80 | 0.12 | < 0.001 |
| contract_monthly [Yes] (vs annual) | 0.90 | 0.20 | < 0.001 |
| visits_per_month (per visit) | −0.30 | 0.05 | < 0.001 |

**1. (3 p)** Compute the predicted probability of cancelling for a member on a monthly contract who visits 2 times per month. Show your working.

**2. (5 p)** (a) Interpret the coefficient −0.30 in terms of odds, and state what is held fixed. (b) A colleague says: "contract_monthly (0.90) matters three times more than visits (−0.30)." Explain why that comparison is not fair.

**3. (6 p)** A manager says: "visits_per_month has p < 0.001, so sending reminder emails that make members visit more will cut cancellations." (a) Is this supported? Explain. (b) Describe an A/B test that could support it: who is randomly assigned, what is compared, and why random assignment matters. (c) Name one thing you would check before trusting the result.

**4. (3 p)** Three models: Model 1 (contract, visits) training AUC 0.74, test 0.73; Model 2 (plus 5 survey measures) 0.81 / 0.79; Model 3 (plus 60 further variables) 0.98 / 0.67. (a) Which model overfits, and how can you tell? (b) Which do you recommend, and why?

## Part C · Cost and threshold (15 points)

A test set has 1,000 members. 120 of them cancelled.

| Threshold | Members flagged | Flagged and cancelled | Not flagged and cancelled | Flagged, did not cancel | Not flagged, did not cancel |
|---|---|---|---|---|---|
| 0.50 | 50 | 36 | 84 | 14 | 866 |
| 0.20 | 180 | 78 | 42 | 102 | 778 |
| 0.10 | 350 | 102 | 18 | 248 | 632 |

**1. (5 p)** At threshold 0.20, compute sensitivity (recall) and precision, and say in one sentence what each means for the chain.

**2. (5 p)** Total cost per 1,000 members: offer nobody 480,000 · offer everybody 440,000 · model at 0.50: 343,000 · model at 0.20: 219,000 · model at 0.10: (to compute). (a) Compute the missing cost. (b) Which policy would you choose, and why? (c) The model at 0.50 is 90.2% accurate and "offer nobody" is 88.0% accurate. What does that comparison hide?

**3. (5 p)** The retention team can contact at most 200 members per 1,000. The CFO says a missed canceller may cost anywhere from 2,000 to 8,000. Costs per 1,000: at 0.50: 175,000 (at 2,000) and 679,000 (at 8,000); at 0.20: 135,000 and 387,000; at 0.10: 160,000 and 268,000. (a) Which thresholds are feasible? (b) Recommend one threshold in two sentences. (c) At a miss cost of 8,000, how much would the chain gain per 1,000 members if it could contact 350?

## Part D · Audit the AI and decide (12 points)

**Item 1 (6 points, 2 each).** Write Supported or Not supported, and give one reason.

- Claim 1: "The model is 89% accurate on the test set, so it is ready to use." (The test cancellation rate is 12%.)
- Claim 2: "At threshold 0.20, the model flags 78 of the 120 members who later cancelled."
- Claim 3: "The AUC is 0.81, so 81% of the members we flag will cancel."

**Item 2 (6 points).** Write the JUDGMENT and ACTION blocks of a board memo, at most 40 words each. They must agree with your numbers and claim no more than the evidence shows. The ACTION must name one number you will monitor after launch.